

A Comparative Analysis of Switched-Current Circuits

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Abstract—Switched Current (SI) samplers are classified as either saturation samplers or triode samplers, based on the operating mode of the sampling transistor. Then the limitations on speed, SNR, and power dissipation are found for the two samplers along with their optimum bias levels. Based on the analysis, an SI sampler using the saturation mode offers the best overall performance. If a sampler based on the triode mode is used, its performance can be maximized with the use of a regulated cascode structure.

I. INTRODUCTION

SINCE their introduction [1]–[3], a wide variety of Switched Current (SI) circuit structures have been proposed. In many cases [1], [4]–[9], design aspects and performance limitations of the individual circuits have been explored. Unfortunately, each circuit configuration has been considered on its own and often in the context of a specific application. Hence, it is difficult to determine which approach offers the best performance for a given application.

The objective of this paper is to determine which SI sampler has the potential to offer the best performance. To begin, it is necessary to classify the various SI circuit structures. Therefore, in Section II, after reviewing the basic operation of an SI sampling circuit, single-ended SI samplers are classified as either saturated samplers or triode samplers. The classification is based on the sampling transistors operating mode. Then, the performance measures of interest; the relative speed, the signal-to-noise ratio and the power dissipation, are defined. These performance measures are then found for the saturation sampler in Section III and for the triode sampler in Section IV. In each section, the performance measures are combined in a figure-of-merit that allows an optimum bias point to be determined. Then, in Section V, the optimum bias point is used to determine the relative advantages of each sampler.

II. SI SAMPLERS

SI samplers are typically built around a single sampling transistor operated in either the saturation or triode mode. The sampling transistor's operating mode determines most of the sampler's characteristics. Consequently, a study of SI samplers is best started by classifying SI samplers based on their mode of operation.

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The simplest SI sampler, illustrated in Fig. 1(a) [2], uses a MOSFET operated in the saturation regime. The sampler has two phases; a sampling phase and a holding phase. During sampling, switches S_1 and S_2 are closed, allowing M1's gate-to-source voltage, V_{gs1} to settle to the level determined by the input current, I_{in} . Clearly, M1 must be in saturation during this operation. Then, during holding, S_1 and S_2 are opened and S_3 is closed, connecting M1 to the output. Provided M1 remains in saturation, the output current I_{out} ideally equals I_{in} . Since M1 operates in saturation, this sampler will be referred to as a saturation sampler.

Variations on the saturation sampler with either a decreased input resistance during sampling or an increased output resistance during holding have been proposed. To lower the input resistance, buffering amplifiers have been used to hold M1's drain-to-source voltage, V_{ds1} constant during sampling [10]. Such an approach creates a virtual ground at the sampler's input, greatly lowering the sampler's input resistance. Alternatively, the samplers output resistance can be increased with either a cascode structure as shown in Fig. 1(b) [11], or a regulated cascode structure, as shown in Fig. 1(c) [12]. By using a level shifter between M2's drain and M1's gate in Fig. 1(b) and (c), M1 can be maintained in the saturation regime and the performance of these two samplers will be similar to that of the saturation sampler, albeit with a decreased input resistance or an increased output resistance. Hence, they may all be considered variations of the saturation sampler.

If M1 is operated in the triode regime, the sampler's behavior differs significantly from that of a saturation sampler. For the cascode case [Fig. 1(b)] V_{ds1} varies with I_{in} . For the regulated cascode case [Fig. 1(c)], V_{ds1} is essentially constant at $V_{D_{S1}}$ due to the additional gain provided by M3. In each case, I_{in} and V_{gs1} are related by the MOSFET's triode equation. Hence, when M1 is operated in triode, these samplers will be referred to as triode samplers.

It is important to distinguish the circuit configuration from its classification as a saturation mode sampler or a triode mode sampler. Indeed, with the use of level shifting, all three samplers shown in Fig. 1 can be operated as either saturation mode samplers or triode mode samplers. If this is done, the finite output resistance of the samplers will differ significantly. Since buffering can be added to any of the circuits to reduce this problem, only the operating mode of M1 remains to determine the circuit's characteristics.

The performance of both the saturation and triode SI samplers can be characterized by a variety of parameters: Foremost

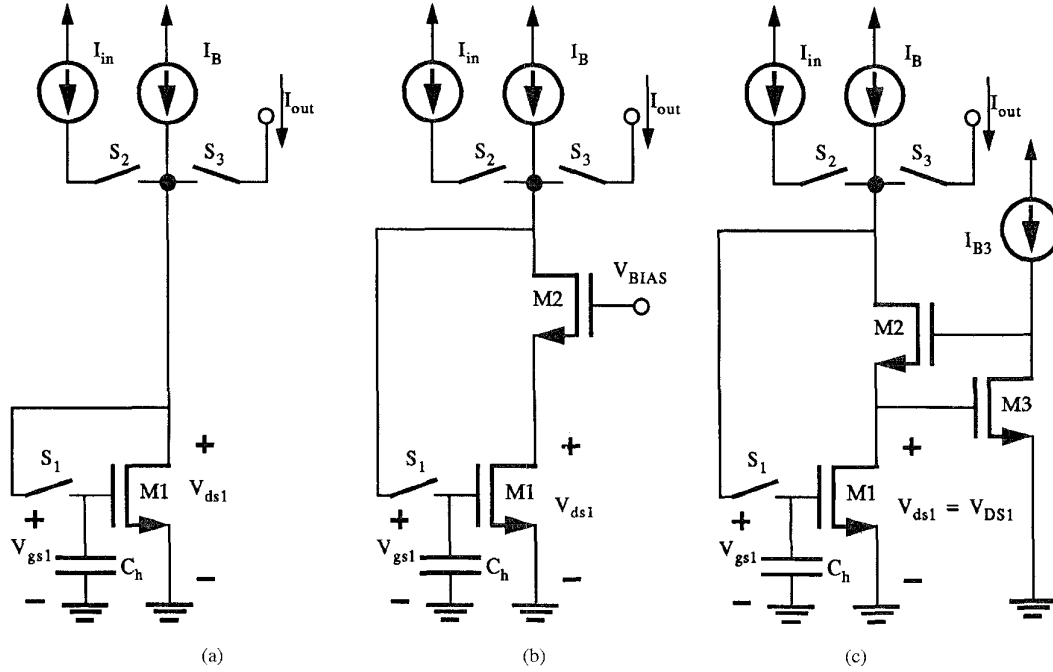


Fig. 1. Common SI current samplers: (a) Saturation sampler. (b) Cascode sampler. (c) Regulated cascode sampler.

among these is the *signal swing*, I_{ss} defined as the allowable range of input currents that can be sampled while maintaining proper circuit operation. Within the allowed signal swing, the accuracy of SI samplers is currently limited by charge injection, provided techniques to reduce the effects of the finite output resistance problem and feedback through the gate-to-drain capacitance are used. As new circuit techniques emerge, such as the SSI technique [13], even the limitations imposed by charge injection will decrease, leaving fundamental limitations such as thermal noise to limit the accuracy. Therefore, the sampler's *signal-to-noise ratio*, SNR, defined as the ratio of the maximum signal power to the worst case noise power represents a fundamental limit on the sampler's accuracy. This accuracy can only be achieved if the sampler settles to the desired accuracy during the sampling phase. If the settling time is insufficient for the desired accuracy, two problems can arise; First, clock jitter will reduce the SNR. Second, if the sampler displays signal dependent settling, harmonic distortion will corrupt the signal. The required settling time is determined by the desired accuracy and the sampler's bandwidth. Hence, the sampler's *minimum bandwidth*, $\omega_{\min}$, can be used to characterize the sampler's speed. Accuracy and speed can only be achieved by dissipating power. Consequently, the sampler's *power dissipation* P_{disp} defined here as the sampler's average current times the supply voltage is of interest. These four parameters, I_{ss} , SNR, $\omega_{\min}$, and P_{disp} each provide useful performance measures.

While individual performance measures are useful, a sampler's overall performance is better characterized with a figure-of-merit that combines the various measures in a single figure. Typically, circuits with a high $\omega_{\min}$, a high SNR and a low P_{disp} are desirable. Hence, a *Figure-Of-Merit*,

FOM, can be defined

$$\text{FOM} = \omega_{\min} \text{SNR} / P_{\text{disp}}. \quad (1)$$

Once the FOM has been found, each sampler's optimum performance can be determined thereby, allowing the relative performance of the two types of samplers to be compared.

To analyze and compare the performance limits of the two samplers, the following assumptions will be made.

- Long channel (i.e., square law) devices are used for M1;
- no channel length modulation (i.e., infinite output resistance in saturation);
- all switches (i.e., S_1 , S_2 , and S_3) have zero on-resistance.

The first assumption is based on the need for a large gate-to-source capacitance to suppress the effects of charge injection. The second assumption can be approximated in practice with buffering circuits to reduce the finite output resistance effects. For the third assumption, wide switches can be used to reduce the switch's on-resistance. The last assumption leads to a first order settling behavior instead of the possibility of critically damped circuits with improved settling times. Unfortunately, some SI samplers exhibit signal dependent parameters, making it difficult to ensure critically damped settling for all signals. Therefore, first order settling yields a more conservative estimate of the sampler's useful speed. Consequently, the following analyzes provides bounds for the performance of the two SI samplers.

III. THE SATURATION SAMPLER

The first sampler type, the saturation sampler, operates with M1 in saturation. Therefore, M1's total input current I_{in} and M1's gate-to-source voltage V_{gs1} are related by

$$I_{\text{in}} = K_1 (V_{gs1} - V_t)^2. \quad (2)$$

K_1 and V_t are M1's conductivity parameter and threshold voltage, respectively. For (2) to be valid, V_{gs1} must exceed V_t to turn M1 on and M1's drain-to-source voltage V_{ds1} can be no less than $V_{gs1} - V_t$ for M1 to remain saturated. Due to the square law relationship of a saturated device, the allowable range of V_{gs1} significantly affects the sampler's performance.

The allowable range of V_{gs1} and hence, the sampler's I_{ss} , is determined by the maximum and minimum useful gate voltages for which M1 remains saturated. The maximum current I_{MAX} in M1 occurs when V_{gs1} is at its maximum $V_{GS\max}$

$$I_{MAX} = K_1(V_{GS\max} - V_t)^2. \quad (3)$$

If V_{ds1} can be increased to $V_{DD} - V_t$, $V_{GS\max}$ can be as high as V_{DD} the supply voltage. The minimum saturation current I_{MIN} according to (2) is zero and occurs when V_{gs1} equals V_t . Hence, (3) represents the saturation sampler's maximum possible I_{ss} .

To take full advantage of the I_{ss} , the sampler must be fast enough to settle during the sampling interval. The saturation sampler is essentially a diode connected MOSFET during sampling. Therefore, its settling time, t_s , for a current step from I_1 to I_2 with an allowable settling error of ε is [14]

$$t_s = \frac{C_h}{2\sqrt{K_1 I_2}} \ln \left[\left(\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}} \right) \left(\frac{\sqrt{I_2 - \varepsilon} + \sqrt{I_2}}{\sqrt{I_2 - \varepsilon} - \sqrt{I_2}} \right) \right] \quad \text{for } I_2 > I_1$$

$$t_s = \frac{C_h}{2\sqrt{K_1 I_2}} \ln \left[\left(\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}} \right) \left(\frac{\sqrt{I_2 + \varepsilon} + \sqrt{I_2}}{\sqrt{I_2 + \varepsilon} - \sqrt{I_2}} \right) \right] \quad \text{for } I_2 < I_1. \quad (4)$$

The settling time depends primarily on I_2 . If ε is very small (4) can be approximated by

$$t_s = \frac{C_h}{2\sqrt{K_1 I_2}} \ln \left[\frac{I_1 - I_2}{\varepsilon} \right]. \quad (5)$$

If I_2 is small, the settling time is very long. Therefore, a bias current, I_B , as shown in Fig. 1(a), is commonly added to avoid long settling times. Typically, to allow bidirectional input currents of I_B is set to

$$I_B = (I_{MAX} - I_{MIN})/2 \quad (6)$$

where I_{MIN} is the current in M1 when I_{in} is at its most negative value (i.e., $I_{in} = I_B - I_{MAX}$). At this point, the sampler's worst case settling time occurs and is given by

$$t_s = \frac{C_h}{2\sqrt{K_1 I_B}} \ln \left[\frac{I_1 - I_{MIN}}{\varepsilon} \right] \quad (7)$$

where $C_h/2\sqrt{K_1 I_{MIN}}$ can be considered the sampler's worst case (i.e., largest) time constant. The inverse of the worst case time constant is the circuit's minimum bandwidth ω_{min}

$$\omega_{min} = 2\sqrt{K_1 I_{MIN}}/C_h. \quad (8)$$

The factor $2\sqrt{K_1 I_{MIN}}$ in (8) is simply M1's minimum transconductance. Hence

$$\omega_{min} = g_{m1(\min)}/C_h \quad (9)$$

can also be used to express the sampler's ω_{min} .

The above analysis is based on the sampler shown in Fig. 1(a) and does not include the effects of a buffering amplifier at the sampler's input. When either a voltage amplifier or a transimpedance amplifier is used at the sampler's input, the sampler's ω_{min} has previously been shown [15] to reduce to (9). Consequently, (8) and its dependence on I_B is typical of any saturation mode sampler.

Using a nonzero I_{MIN} improves the sampler's speed but, reduces the sampler's signal swing. When I_{MIN} is not zero, the sampler's I_{ss} becomes

$$I_{ss} = I_{MAX} - I_{MIN}. \quad (10)$$

Therefore, the saturation sampler has a tradeoff between its speed and signal swing.

Along with reducing I_{ss} , a nonzero I_{MIN} increases the sampler's P_{disp} . The saturation sampler's I_{disp} , if all signal levels are assumed to be equally likely, can be expressed as

$$P_{disp} = V_{DD} \left(I_{MIN} + \frac{I_{ss}}{2} \right). \quad (11)$$

A zero I_{MIN} minimizes P_{disp} and maximizes I_{ss} but also yields an unacceptable ω_{min} . The variation of these parameters for different levels of I_{MIN} is illustrated in Fig. 2.

The last parameter to consider is the sampler's thermal noise and the resulting SNR. The saturation sampler's thermal noise arises from the device used to generate I_B , M1's channel noise, and the on-resistance of switches S_1 , S_2 , and S_3 . I_B can be generated with a variety of different circuits and biasing arrangements. Hence, the noise contribution due to I_B is unknown, unless a specific circuit is being considered. If it is assumed that the circuit used to generate I_B is an ideal current source, the resulting SNR will be an upper limit on the achievable SNR. Within the sampler itself, the dominant noise, during both the sampling and holding phases is M1's channel noise [16].

For the saturation sampler, M1 is operated in saturation. Therefore, M1's mean squared noise current $\overline{i_{device}^2}$ is given by

$$\overline{i_{device}^2} = 4kT^2 g_{m1} \Delta f \quad (12)$$

where k is Boltzmann's constant, T is the absolute temperature and Δf is the noise bandwidth. Assuming the sampler has sufficient time to settle, the noise bandwidth is

$$\Delta f = \frac{g_{m1}}{4C_h}. \quad (13)$$

This noise bandwidth leads to a mean squared sampled noise current $\overline{i_s^2}$ of

$$\overline{i_s^2} = g_{m1}^2 \left(\frac{2}{3} \frac{kT}{C_h} \right) \quad (14)$$

which degrades the accuracy of the sampled current.

During the holding phase, I_{out} consists of the memorized input current, the memorized noise current and the sampler's thermal noise current. Again the dominant noise source is M1's thermal noise current, as given by (12). To determine Δf it is reasonable to assume that the sampler drives an identical sampler. Therefore, the noise bandwidth will again be given

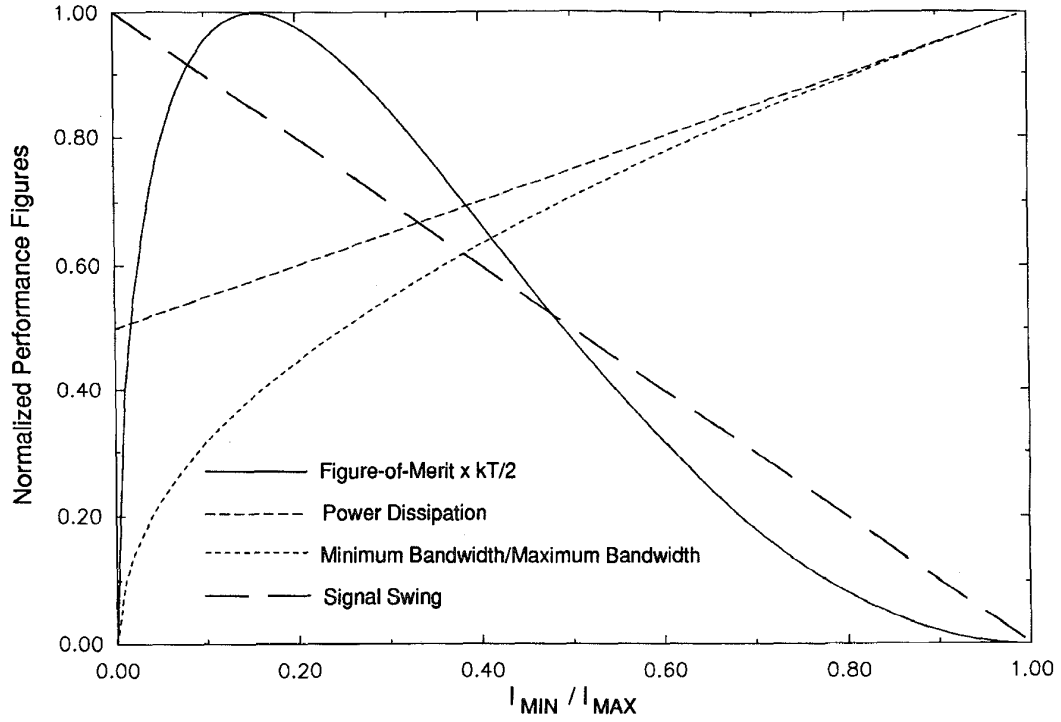


Fig. 2. Normalized performance measures for the saturation sampler. Normalized to $K = 1$, $V_{DD} = 1$, $V_t = 0.2V_{DD}$, and $C_h = 1$.

by (13). Consequently, the sampler generates a mean squared noise current during the hold period of

$$\overline{i_h^2} = g_{m1}^2 \left(\frac{2}{3} \frac{kT}{C_h} \right). \quad (15)$$

Provided a sufficient settling time for high accuracy applications is used, the noise memorized during sampling and that generated during holding are uncorrelated. Therefore, the sampler's total mean squared noise current $\overline{i_{total}^2}$ is

$$\begin{aligned} \overline{i_{total}^2} &= \overline{i_s^2} + \overline{i_h^2} \\ &= g_{m1}^2 \left(\frac{4}{3} \frac{kT}{C_h} \right). \end{aligned} \quad (16)$$

The worst case $\overline{i_{total}^2}$ occurs when g_{m1} is at its maximum

$$g_{m1(max)} = 2\sqrt{K_1 I_{MAX}}. \quad (17)$$

Assuming a sinusoidal signal with a peak-to-peak swing of I_{ss} , the saturation sampler's SNR can be expressed as

$$\begin{aligned} \text{SNR} &= \frac{(I_{ss}/2\sqrt{2})^2}{g_{m1(max)}^2 (4kT/3C_h)} \\ &= \frac{3(I_{MAX} - I_B)^2}{32K_1 I_{MAX}} \left(\frac{C_h}{kT} \right). \end{aligned} \quad (18)$$

The saturation sampler's SNR is maximized by setting I_{MIN} to zero.

With the various performance measures determined, the saturation sampler's figure-of-merit can be found. The FOM is found by combining (8), (11), and (18)

$$\text{FOM} = \frac{3\sqrt{I_B}(I_{MAX} - I_{MIN})^2}{8kTV_{DD}\sqrt{K_1}I_{MAX}(I_{MAX} - I_{MIN})}. \quad (19)$$

The I_{MIN} and corresponding I_B that optimizes the FOM, $I_{MIN(opt)}$ and $I_{B(opt)}$ are

$$\begin{aligned} I_{MIN(opt)} &= 0.1547I_{MAX} \\ I_{B(opt)} &= 0.58I_{MAX}. \end{aligned} \quad (20)$$

An alternative view of I_{MAX} , I_{MIN} and I_B , is that the sampler is biased with a current of I_B . Then a modulation index [5], [6], m , defined as

$$m = (I_{MAX} - I_{MIN}) / (I_{MAX} + I_{MIN}) \quad (21)$$

is used to specify I_{MAX} and I_{MIN} . Using this approach, the optimum value for m is 0.73. A summary of the saturation sampler's performance measures and their values when $I_B = I_{B(opt)}$ is shown in Table I.

Based on the above analysis and Table I, the saturation sampler's performance is largely determined by the signal level. Both the sampler's noise, as given by (16), and its minimum bandwidth, as given by (8), are signal dependent. The worst case noise and bandwidth occur at the maximum and minimum signal levels, respectively. Hence they are largely independent parameters.

The worst case noise is determined by M1's I_{MAX} and is unaffected by M1's I_{MIN} . As shown by (18), increasing I_{MAX} , which implies increasing $V_{GS\max}$, increases the signal power faster than the noise power, yielding an improved SNR. Provided the resulting $V_{GS\max}$ is less than V_{DD} , increasing $V_{GS\max}$ leads to an almost 1:1 SNR to P_{disp} tradeoff. Consequently, increasing the saturation sampler's $V_{GS\max}$ to V_{DD} yields the theoretical maximum SNR.

TABLE I
PERFORMANCE SUMMARY OF THE SATURATION SAMPLER

Parameter	General Expression (Voltage Based)	General Expression (Current Based)	Optimum Value ($V_{GSmax} = V_{DD}$)
Signal Swing (I_{ss})	$K_1 [(V_{GSmax} - V_t)^2 - (V_{GSmin} - V_t)^2]$	$I_{MAX} - I_{MIN}$	$0.85K_1 (V_{DD} - V_t)^2$
Minimum Bandwidth (ω_{min})	$\frac{2K_1 (V_{GSmin} - V_t)}{C_h}$	$\frac{2\sqrt{K_1 I_{MIN}}}{C_h}$	$0.78 \frac{K_1 (V_{DD} - V_t)}{C_h}$
Power Dissipation (P_{disp})	$\frac{K_1 V_{DD}}{2} [(V_{GSmax} - V_t)^2 + (V_{GSmin} - V_t)^2]$	$\frac{V_{DD}}{2} (I_{MAX} + I_{MIN})$	$0.58K_1 V_{DD} (V_{DD} - V_t)^2$
Signal-to-Noise Ratio (SNR)	$\frac{3C_h [(V_{GSmax} - V_t)^2 - (V_{GSmin} - V_t)^2]^2}{128kT (V_{GSmax} - V_t)^2}$	$\frac{3C_h (I_{MAX} - I_{MIN})^2}{128kTK_1 I_{MAX}}$	$\frac{0.017C_h}{kT} (V_{DD} - V_t)^2$
Figure-of-Merit (FOM)	$\frac{3(V_{GSmin} - V_t) [(V_{GSmax} - V_t)^2 - (V_{GSmin} - V_t)^2]^2}{32kTV_{DD} (V_{GSmax} - V_t)^2 [(V_{GSmax} - V_t)^2 + (V_{GSmin} - V_t)^2]}$	$\frac{3\sqrt{I_B} (I_{MAX} - I_{MIN})^2}{32kTV_{DD} \sqrt{K_1} I_{MAX} (I_{MAX} + I_{MIN})}$	$\frac{0.023}{kT} \left(\frac{V_{DD} - V_t}{V_{DD}} \right)$
Optimum Bias	$V_{GSmin} - V_t = 0.39 (V_{GSmax} - V_t)$	$I_B = 0.58I_{MAX}$	

To estimate a saturation sampler's maximum SNR, when biased for the optimum FOM, values of $V_{DD} = 5$ V, $V_t = 0.2V_{DD}$, $C_h = 1$ pF, and $T = 300$ K° can be used. This yields an SNR of 78 dB which, is higher than the 67 dB SNR recently reported for a high performance SI circuit [17]. Hence, as new techniques arise, one can expect further SNR improvements. Nevertheless, unless thermal noise can be avoided or reduced, there is not much to be gained by reducing other noise sources below the level indicated by (18).

At the other extreme, the sampler's ω_{min} is determined by M1's minimum allowed gate voltage which sets I_{MIN} . Hence, the use of an I_B greater than half I_{MAX} increases the sampler's speed due to the increased I_{MIN} . For small increases in I_{MIN} , both the SNR and P_{disp} remain largely unaffected. Consequently, the use of a small I_B such as $I_B = I_{B(opt)}$ yields a significant speed increase without significantly degrading the saturation sampler's other performance measures.

IV. THE TRIODE SAMPLER

The second sampler type is based on triode mode operation. For a triode mode sampler, M1's total current, I_{in} can be expressed as

$$I_{in} = K_1(2(V_{gs1} - V_t)V_{ds1} - V_{ds1}^2). \quad (22)$$

To remain in triode, V_{gs1} must exceed V_{ds1} by at least V_t . The level of V_{ds1} is determined by the source voltage of M2 in Fig. 1(b) or the gate voltage of M3 in Fig. 1(c). Since M1 operates in triode, M2 should be operated in saturation to provide a high output resistance for the sampler. Therefore, I_{in} can also be expressed as

$$I_{in} = K_2(V_{gs2} - V_t)^2 \quad (23)$$

where V_{gs2} is the difference between the bias voltage, V_{BIAS} and V_{ds1} ,

$$V_{gs2} = V_{BIAS} - V_{ds1} \quad (24)$$

therefore

$$I_{in} = K_2(V_{BIAS} - V_{ds1} - V_t)^2. \quad (25)$$

Based on (22) and (25), the operation of a triode sampler and the sampler's performance depends on both the biasing of M2 and the relative sizes of M1 and M2.

The triode sampler's signal swing is determined by the minimum gate voltage for which M1 remains in triode and by the maximum gate voltage for M1 allowed by the supply voltage. M1's minimum gate voltage, for triode mode operation occurs when

$$V_{gs1} = V_{ds1} + V_t \quad (26)$$

substituting (26) into (22) and (25) yields the minimum gate voltage V_{gsmin} for triode operation

$$V_{GSmin} = \frac{\sqrt{K_2/K_1}V_{bias} + V + t}{1 + \sqrt{K_2/K_1}}. \quad (27)$$

K_2/K_1 is the ratio of the size of M2 to that of M1 and will be referred to as the K ratio

$$K = K_2/K_1. \quad (28)$$

Knowing V_{GSmin} , the minimum triode current I_{MIN} can be expressed as

$$I_{MIN} = \frac{K_1 K}{(1 + \sqrt{K})^2} (V_{BIAS} - V_t)^2. \quad (29)$$

If I_{in} increases, V_{ds1} and V_{gs1} will change. Based on (22) and (25), V_{ds1} decreases while V_{gs1} increases. A plot illustrating the variation in V_{ds1} and I_{in} (assuming $I_B = 0$) versus V_{gs1} in the triode mode is shown in Fig. 3. Note that V_{ds1} is constant when the K ratio is infinite. When V_{gs1} equals V_{DD} , the maximum current occurs. While I_{MAX} can be found by solving (22) and (25) for $V_{gs1} = DD$, the result is not particularly useful due to the number of parameters involved.

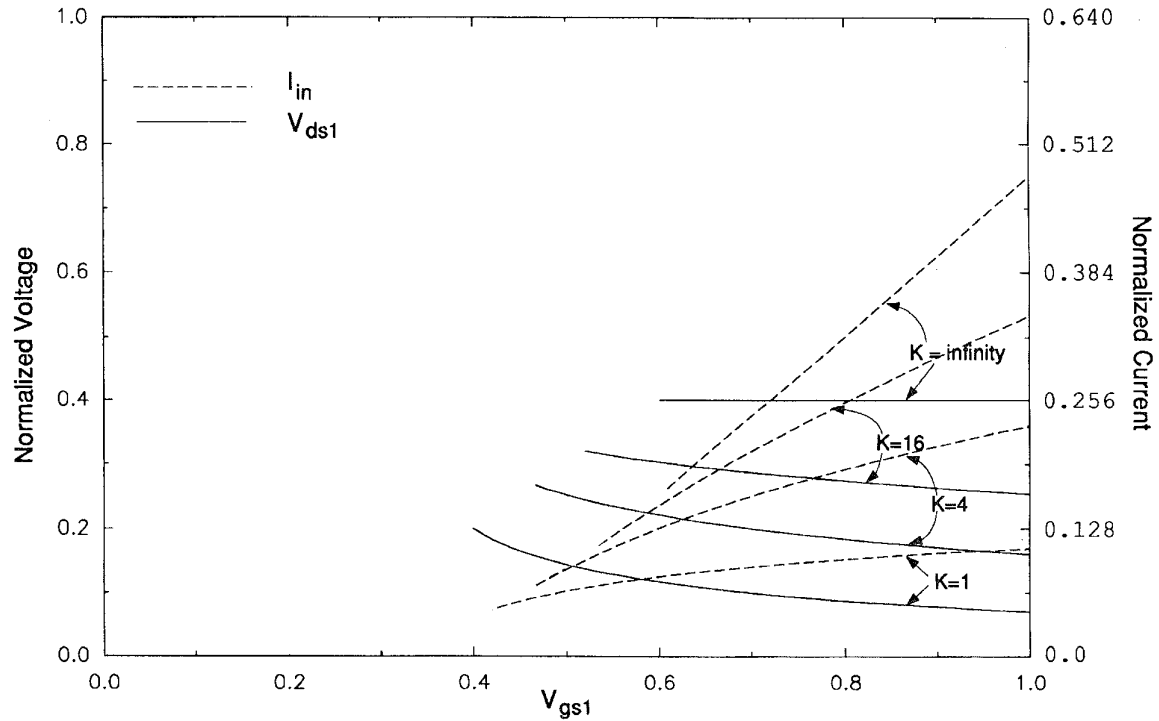


Fig. 3. A normalized plot of V_{ds1} and I_{in} versus V_{gs1} for the triode sampler with $V_{BIAS} = 0.6V_{DD}$. $K = 1$, $V_{DD} = 1$, and $V_t = 0.2V_{DD}$ were used to normalize the plot.

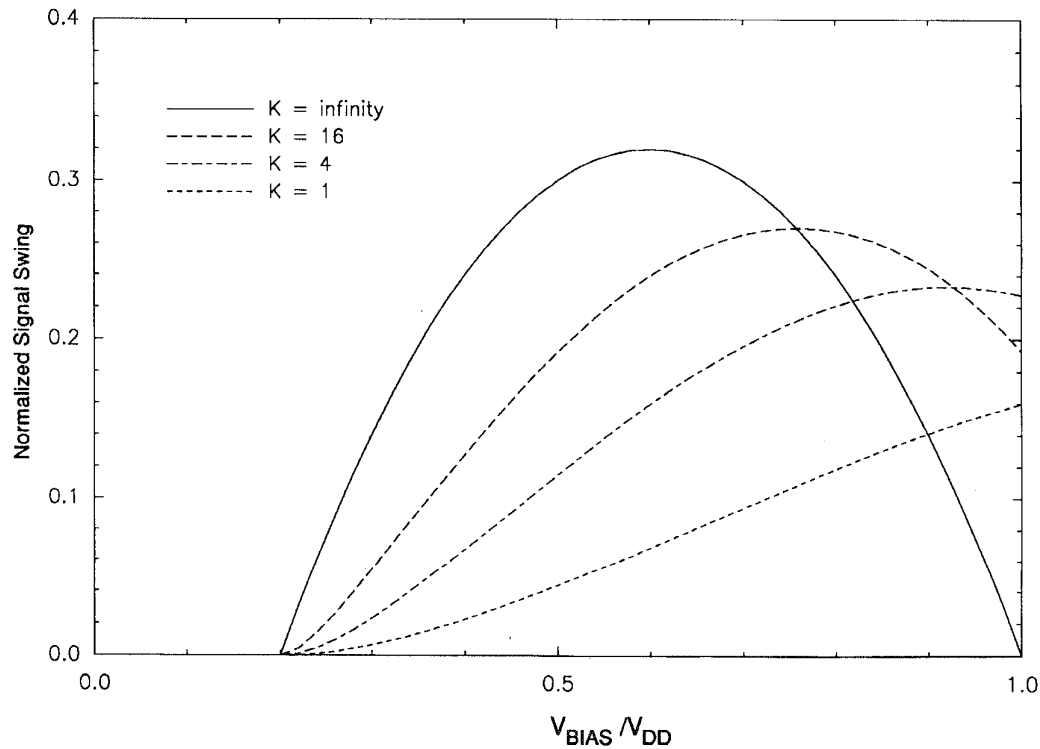


Fig. 4. Normalized signal swing versus V_{BIAS}/V_{DD} for the triode sampler. $K_1 = 1$, $V_{DD} = 1$ and $V_t = 0.2V_{DD}$ were used to normalize the plot.

More insight can be obtained by using SPICE simulations to generate the normalized plot of the sampler's signal swing shown in Fig. 4. The effects of the K ratio and V_{BIAS} on I_{MIN} , I_{MAX} , and I_{ss} are clearly evident.

The maximum I_{ss} occurs when K is infinite. While not possible, an infinite K ratio can be closely approximated with the use of a regulated cascode sampler. For the regulated cascode, I_{MIN} occurs when M1 is at the edge of saturation

((26)) leading to

$$I_{\text{MIN}} = K_1 V_{DS1}^2. \quad (30)$$

The maximum current occurs when V_{gs1} is at its maximum, $V_{GS \text{ max}}$

$$I_{\text{MAX}} = K_1 (2(V_{GS \text{ max}} - V_t)V_{DS1} - V_{DS1}^2). \quad (31)$$

$V_{GS \text{ max}}$ can be as high as V_{DD} . Combining (30) and (31) yields the maximum signal swing for any V_{DS1}

$$\begin{aligned} I_{ss} &= I_{\text{MAX}} - I_{\text{MIN}} \\ &= 2K_1 ((V_{GS \text{ max}} - V_t)V_{DS1} - V_{DS1}^2) \end{aligned} \quad (32)$$

I_{ss} is a quadratic in V_{DS1} with its maximum occurring at

$$V_{DS1} = (V_{GS \text{ max}} - V_t)/2. \quad (33)$$

Since the K ratio is infinite, this V_{DS1} corresponds to an effective V_{BIAS} of

$$\begin{aligned} V_{\text{BIAS}} &= V_{DS1} + V_t \\ &= \frac{1}{2}(V_{GS \text{ max}} + V_t). \end{aligned} \quad (34)$$

Similar to the saturation sampler though, it will be found that maximizing I_{ss} does not optimize the sampler's overall performance.

To characterize the triode sampler's speed, a settling time expression based on the small signal model can be obtained. However, due to the numerous parameters involved, such an approach does not yield much insight into the circuit's operation. Alternatively, since it is known that C_h is typically the circuit's largest capacitance, the triode sampler's dominant pole, ω_{dom} can be estimated as

$$\omega_{\text{dom}} = g_{m1}/C_h. \quad (35)$$

The validity of this estimate can then be determined with SPICE simulations. As discussed below, the regulated cascode's ω_{dom} is higher than that of the cascode sampler's. Therefore, only simulation results for the regulated cascode sampler [Fig. 1(c)] will be presented here. Initially, the ratio of g_{m1}/C_h to the circuit's dominant pole for various widths of M1 and values of C_h was determined. The results are shown in Fig. 5(a). For the simulations, the W/L of M2 and M3 were $10 \mu\text{m}/1.2 \mu\text{m}$ and $4 \mu\text{m}/1.2 \mu\text{m}$, respectively, while M1 had a $1.2 \mu\text{m}$ channel length. Based on these dimensions, the gate-to-source capacitances of M1, M2, and M3 are in the 10 to 20 fF range. As shown in Fig. 5(a), for C_h much larger than the gate-to-source capacitances, g_{m1}/C_h is a good estimate of the dominant pole. Indeed, for C_h greater than 0.4 pF, the approximation introduces an error of less than 20% in ω_{dom} and is largely unaffected by changes in the width of M1 if C_h includes M1's gate-to-source capacitance. Then, the effect of changing the amplifier, M3 was considered. As shown in Fig. 5(b), even large changes in the ratio K_3/K_1 do not significantly affect the validity of the dominant pole

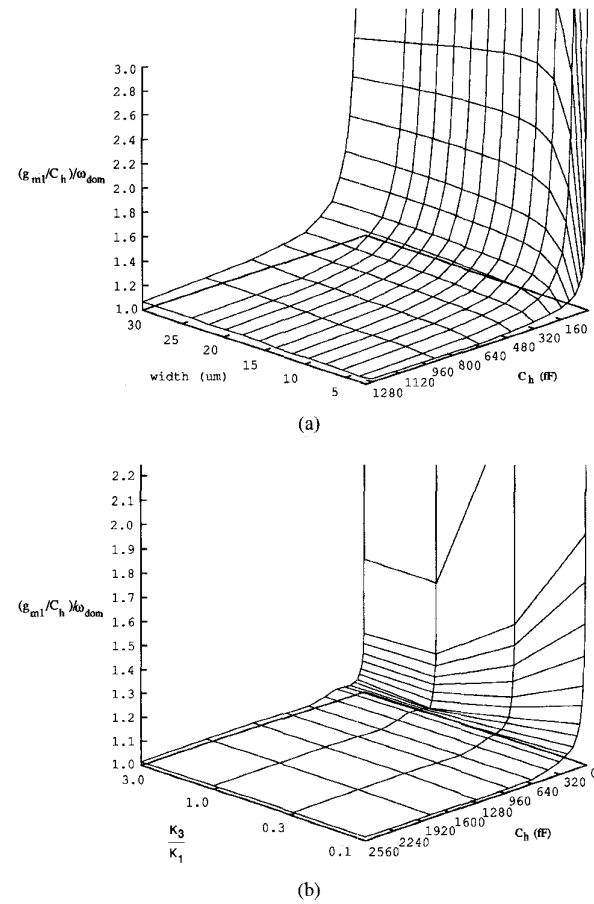


Fig. 5. (a) Variation of $(g_{m1}/C_h)/\omega_{\text{dom}}$ versus C_h and the width of M1 for the regulated cascode sampler. (b) Variation of $(g_{m1}/C_h)/\omega_{\text{dom}}$ versus C_h and K_3/K_1 for the regulated cascode sampler.

approximation. Consequently, (35) is a good estimate of the triode sampler's dominant pole.

The ω_{dom} given in (35) depends on g_{m1} . In triode, g_{m1} is given by

$$g_{m1} = 2K_1 V_{ds1}. \quad (36)$$

For a fixed bias level, V_{ds1} increases with the K ratio, as shown in Fig. 3. Therefore, the maximum V_{ds1} and hence the maximum speed are achieved when the K ratio is infinite. Consequently, the triode sampler's best, worst case dominant pole is

$$\omega_{\text{in}} = \frac{2K_1 V_{DS1}}{C_h}. \quad (37)$$

Since V_{DS1} is constant when the K ratio is infinite, ω_{min} is signal independent.

Increasing V_{DS1} increases the sampler's speed, but also increases the sampler's P_{disp} . The triode sampler's P_{disp} is composed of two parts for the cascode sampler and three parts for the regulated cascode sampler. For both samplers, the signal current, represented by I_{ss} , and the I_B necessary to maintain triode operation dissipate power. For the regulated cascode sampler, the amplifier also dissipates power. As shown in Fig. 5(b), K_3 (and hence its bias current) can be made

small with respect to K_1 without affecting the circuit's speed. Consequently, the amplifier's bias current can be neglected and P_{disp} will be dominated by I_{ss} and I_{MIN} or I_B

$$\begin{aligned} P_{\text{disp}} &= V_{DD} I_B \\ &= K_1 V_{DD} (V_{GS\text{max}} - V_t) V_{DS1}. \end{aligned} \quad (38)$$

Comparing (38) and (37) illustrates that the triode sampler has a speed-power tradeoff.

Before optimizing the triode sampler's performance, the sampler's thermal noise and the resulting SNR should be found. For the triode sampler, thermal noise arises from many different sources. Like the saturation sampler though, the dominant noise source is M1's channel noise.

When M1 is operated in triode, its mean squared channel noise current is given by

$$\overline{i_{\text{device}}^2} = \frac{4kT\Delta f}{R_{\text{on}}} \quad (39)$$

where R_{on} is M1's on-resistance. During sampling, the noise bandwidth will be

$$\Delta f = \frac{g_{m1}}{4C_h}. \quad (40)$$

Therefore, the mean squared sampled noise current $\overline{i_s^2}$ will be

$$\overline{i_s^2} = \frac{g_{m1}}{R_{\text{on}}} \frac{kT}{C_h}. \quad (41)$$

During the holding phase, if the output is sampled by a similar sampler, the noise bandwidth will also be given by (40). Hence, the channel current's mean squared noise current during the hold phase $\overline{i_h^2}$ will be

$$\overline{i_h^2} = \frac{g_{m1}}{R_{\text{on}}} \frac{kT}{C_h}. \quad (42)$$

The sampler's output current then consists of the desired current along with the two thermal noise components. Combining (41) and (42) yields the total mean squared noise current at the sampler's output

$$\begin{aligned} \overline{i_{\text{total}}^2} &= \overline{i_s^2} + \overline{i_h^2} \\ &= \frac{2g_{m1}}{R_{\text{on}}} \frac{kT}{C_h}. \end{aligned} \quad (43)$$

For the triode mode, R_{on} is given by

$$R_{\text{on}} = [2K_1(V_{gs} - V_t)]^{-1} \quad (44)$$

while g_{m1} is given by

$$g_{m1} = 2K_1 V_{ds1}. \quad (45)$$

Substituting for R_{on} and g_{m1} in (43) yields

$$\overline{i_{\text{total}}^2} = 8K_1^2 (V_{gs1} - V_t) V_{ds1} kT / C_h. \quad (46)$$

The worst case total noise occurs when V_{gs1} is equal to $V_{GS\text{max}}$. Assuming the signal is a sinusoid with a peak-to-peak amplitude of I_{ss} , the signal power (i.e., $(I_{ss}/2\sqrt{2})^2$) divided by $\overline{i_{\text{total}}^2}$ yields the triode sampler's SNR which is

illustrated in Fig. 6. The maximum SNR is obtained when the K ratio is infinite. For this case, the SNR can be expressed as

$$\text{SNR} = \frac{[(V_{GS\text{max}} - V_t)V_{DS1} - V_{DS1}^2]^2}{16(V_{GS\text{max}} - V_t)V_{DS1}} \left[\frac{C_h}{kT} \right]. \quad (47)$$

The SNR is maximized by setting V_{DS1} to

$$V_{DS1(\text{opt})} = (V_{GS\text{max}} - V_t)/3 \quad (48)$$

which leads to a maximum SNR of

$$\text{SNR} = 0.0093 C_h (V_{GS\text{max}} - V_t)^2 / kTs \quad (49)$$

for the triode sampler.

To complete the triode sampler's analysis, its FOM must be found. The FOM is found by multiplying ω_{min} (37) by the SNR (47) and dividing by P_{disp} (38) to obtain

$$\text{FOM} = \frac{[(V_{GS\text{max}} - V_t)V_{DS1} - V_{DS1}^2]^2}{8V_{DD}kT(V_{GS\text{max}} - V_t)^2V_{DS1}}. \quad (50)$$

As illustrated in Fig. 7, the FOM is maximized by an infinite K ratio. The optimum bias point for the FOM is given by $V_{DS1(\text{opt})}$ (48) that also maximizes the SNR.

A summary of the triode sampler's performance measures and their values when an infinite K ratio and the optimum bias point are used is shown in Table II. Both the voltage based and the current based expressions are given to facilitate comparisons. For the triode mode sampler's current based expressions, it is important to note that I_{MAX} is a function of I_{MIN}

$$I_{\text{MAX}} = 2(V_{DD} - V_t)\sqrt{K_1 I_{\text{MIN}}} - I_{\text{MIN}}. \quad (51)$$

Making this substitution for I_{MAX} yields the optimum values shown in Table II when the sampler is biased for its optimum performance.

Based on the above analysis, the triode sampler's performance is largely determined by its K ratio and bias level. Increasing the K ratio reduces the signal dependent variations in V_{ds1} , leading to an increase in the minimum speed, the signal swing, and overall performance for a given bias level. Indeed, if a large enough K ratio is used, the sampler's speed becomes signal independent. The remaining factor, the bias level, affects both the sampler's speed and its worst case thermal noise. Consequently, the selection of an appropriate bias level for the triode sampler always implies an optimization.

V. DISCUSSION

The results of Sections III and VI can now be used to compare the behavior and overall performance of the two general classes of SI samplers. In general, the operating behavior of the saturation sampler is largely signal dependent while the triode sampler's behavior is largely bias dependent.

For the saturation sampler, signal dependent operation leads to a signal dependent settling time. Consequently, small currents should be avoided to prevent long settling times. On the other hand, the saturation sampler's worst case noise is only determined by the maximum signal

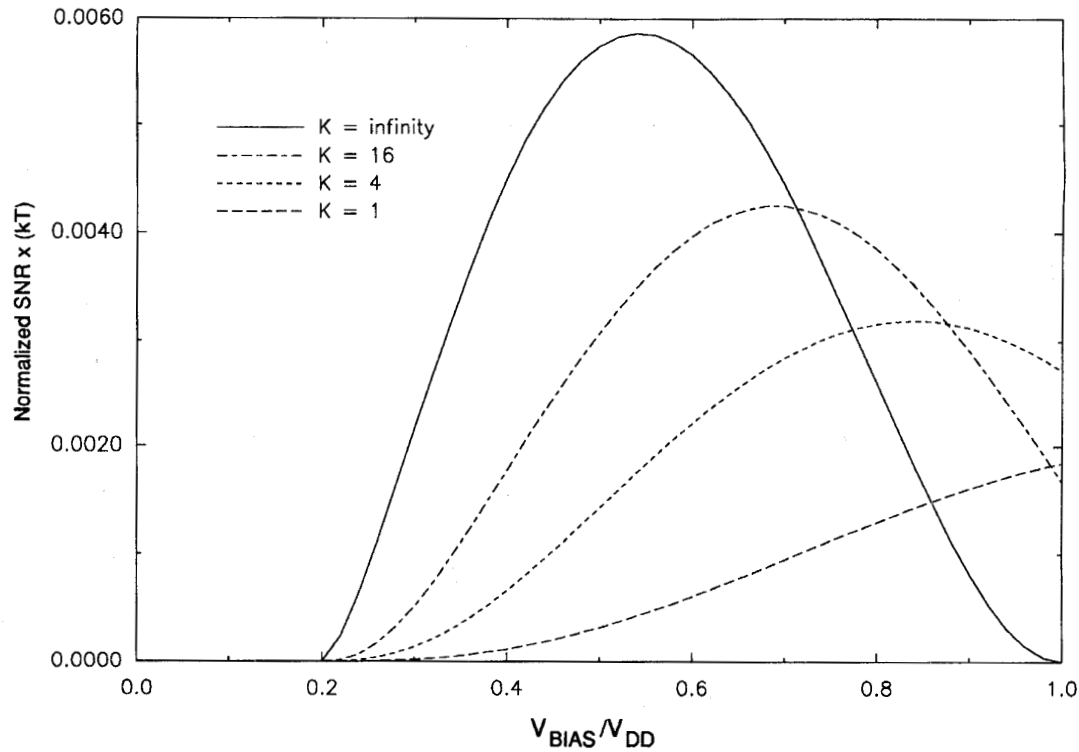


Fig. 6. Normalized $\text{SNR} \times (kT)$ versus $V_{\text{BIAS}}/V_{\text{DD}}$ for the triode sampler. $K_1 = 1$, $V_{\text{DD}} = 1$, $V_t = 0.2V_{\text{DD}}$, and $C_h = 1$ were used to normalize the plot.

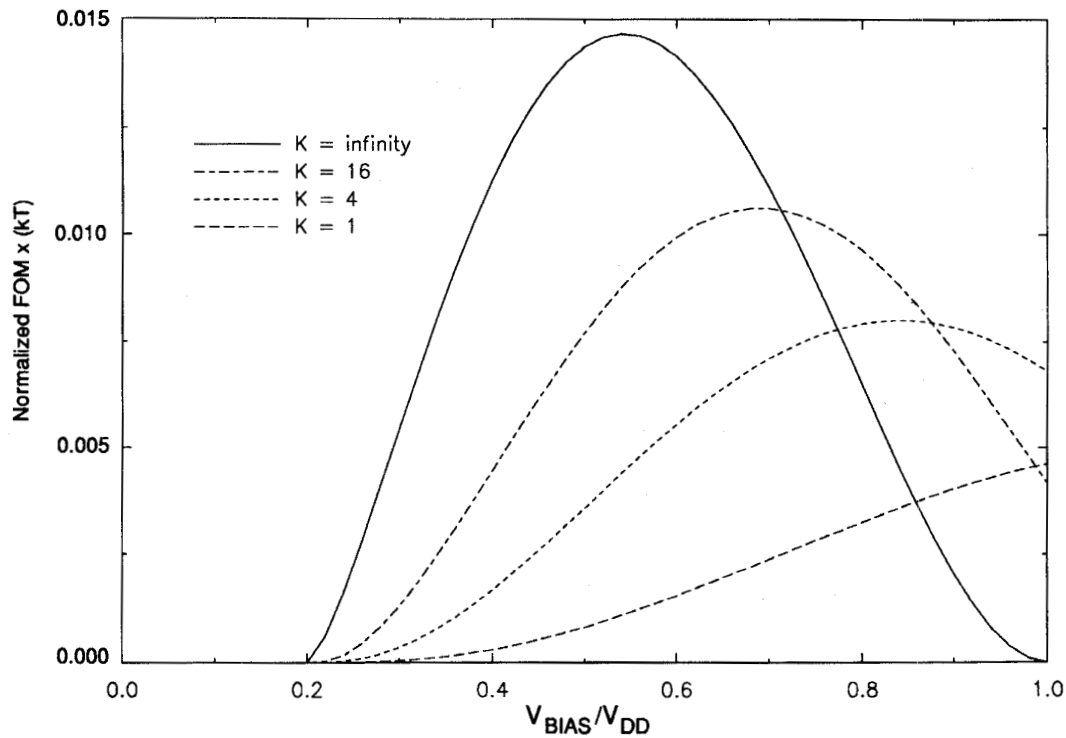


Fig. 7. Normalized $\text{FOM} \times (kT)$ versus $V_{\text{BIAS}}/V_{\text{DD}}$ for the triode sampler. $K_1 = 1$, $V_{\text{DD}} = 1$, $V_t = 0.2V_{\text{DD}}$, and $C_h = 1$ were used to normalize the plot.

level. Hence, the sampler's speed can be altered without significantly affecting the noise level.

For the triode sampler, the situation is noticeably different. If a regulated cascode configuration is used, the sampler's

speed is determined primarily by the bias level and is largely unaffected by the signal level. Unfortunately, the triode sampler's noise level also depends on the bias level. Therefore, selecting a bias level leads to a tradeoff between speed

TABLE II

Parameter	General Expression (Voltage Based)	General Expression (Current Based)	Optimum Value ($V_{GSmax} = V_{DD}$)
Signal Swing (I_{ss})	$2K_1 [(V_{GSmax} - V_t) V_{DS1} - V_{DS1}^2]$	$I_{MAX} - I_{MIN}$	$0.44K_1 (V_{DD} - V_t)^2$
Minimum Bandwidth (ω_{min})	$\frac{2K_1 V_{DS1}}{C_h}$	$\frac{2\sqrt{K_1 I_{MIN}}}{C_h}$	$0.67 \frac{K_1 (V_{DD} - V_t)}{C_h}$
Power Dissipation (P_{disp})	$K_1 V_{DD} (V_{GSmax} - V_t) V_{DS1}$	$\frac{V_{DD}}{2} (I_{MAX} + I_{MIN})$	$0.33K_1 V_{DD} (V_{DD} - V_t)^2$
Signal-to-Noise Ratio (SNR)	$\frac{C_h [(V_{GSmax} - V_t) V_{DS1} - V_{DS1}^2]^2}{16kT(V_{GSmax} - V_t) V_{DS1}}$	$\frac{C_h (I_{MAX} - I_{MIN})^2}{32kTK_1 (I_{MAX} + I_{MIN})}$	$\frac{0.0093C_h}{kT} (V_{DD} - V_t)^2$
Figure-of-Merit (FOM)	$\frac{[(V_{GSmax} - V_t) V_{DS1} - V_{DS1}^2]^2}{8kTV_{DD} (V_{GSmax} - V_t)^2 V_{DS1}}$	$\frac{\sqrt{I_{MIN}}}{8kTV_{DD} \sqrt{K_1}} \left[\frac{I_{MAX} - I_{MIN}}{I_{MAX} + I_{MIN}} \right]^2$	$\frac{0.019}{kT} \left(\frac{V_{DD} - V_t}{V_{DD}} \right)$
Optimum Bias ($K = \infty$)	$V_{DS1} = \frac{V_{GSmax} - V_t}{3}$	$I_B = 0.60I_{MAX}$	

and SNR. Based on this observation, one would expect the saturation sampler to out-perform the triode sampler.

When the FOM of the saturation and triode samplers are both maximized, the saturation sampler out-performs the triode sampler. Indeed, at the optimum bias, the saturation sampler provides a larger signal swing, a higher ω_{min} and a higher SNR. Consequently, the saturation sampler is preferred for high performance applications.

The above conclusions are based on optimizing the speed, SNR, and P_{disp} . When P_{disp} is not an issue, an alternate FOM, FOM_{alt}, proposed earlier [5], [6], can be used

$$\text{FOM}_{alt} = \omega_{min} \text{SNR}. \quad (52)$$

For the saturation sampler, FOM_{alt} is given by

$$\begin{aligned} \text{FOM}_{alt} &= \frac{3K_1 (V_{GSmin} - V_t) [(V_{GSmax} - V_t)^2 - (V_{GSmin} - V_t)^2]}{64kT(V_{GSmax} - V_t)^2} \end{aligned} \quad (53)$$

and is optimized by

$$V_{GSmin} - V_t = (V_{GSmax} - V_t)/\sqrt{5}. \quad (54)$$

For the triode sampler, FOM_{alt} is given by

$$\text{FOM}_{alt} = \frac{K_1 [(V_{GSmax} - V_t) V_{DS1} - V_{DS1}^2]^2}{8kT(V_{GSmax} - V_t)} \quad (55)$$

and is optimized by

$$V_{DS1} = (V_{GSmax} - V_t)/2. \quad (56)$$

The resulting performance parameters for the two samplers are shown in Table III. For both samplers, this optimization yields a higher ω_{min} and a lower SNR. For the triode sampler, this bias level corresponds to maximizing the signal swing. Once

TABLE III

Parameter	Saturation Sampler ($V_{GSmax} = V_{DD}$)	Triode Sampler ($V_{GSmax} = V_{DD}$, $K = \infty$)
Signal Swing (I_{ss})	$0.80K_1 (V_{DD} - V_t)^2$	$0.50K_1 (V_{DD} - V_t)^2$
Minimum Bandwidth (ω_{min})	$0.89 \frac{K_1 (V_{DD} - V_t)}{C_h}$	$1.00 \frac{K_1 (V_{DD} - V_t)}{C_h}$
Power Dissipation (P_{disp})	$0.60K_1 V_{DD} (V_{DD} - V_t)^2$	$0.50K_1 V_{DD} (V_{DD} - V_t)^2$
Signal-to-Noise Ratio (SNR)	$\frac{0.015C_h}{kT} (V_{DD} - V_t)^2$	$\frac{0.0078C_h}{kT} (V_{DD} - V_t)^2$
Figure-of-Merit (FOM)	$\frac{0.022}{kT} \left(\frac{V_{DD} - V_t}{V_{DD}} \right)$	$\frac{0.016}{kT} \left(\frac{V_{DD} - V_t}{V_{DD}} \right)$
Alt. Figure-of-Merit (FOM _{alt})	$\frac{0.013K_1}{kT} (V_{DD} - V_t)^3$	$\frac{0.0078K_1}{kT} (V_{DD} - V_t)^3$

again, the results indicate that the saturation sampler yields better overall performance.

Although the two sampling modes lead to performance differences, the analysis also reveals that the performance of the two samplers are primarily controlled by the same parameters. To maximize the signal swing and SNR of both samplers, it is desirable to use the highest possible V_{GSmax} . Once V_{GSmax} is known, the optimum V_{GSmin} can be determined. For both samplers a V_{GSmin} greater than V_t is required to avoid long settling times. Once V_{GSmax} and V_{GSmin} have been found, the performance equations for the two samplers are of the same form.

For a given V_{GSmax} and V_{GSmin} , a sampler's FOM is fixed. Within this constraint, one can use K_1 and C_h to tradeoff one performance parameter for another. K_1 can be used to tradeoff P_{disp} and speed while C_h can be used to tradeoff SNR and speed. Consequently, both samplers can be tailored to specific applications.

While it is convenient to consider K_1 and C_h separately for the analysis, C_h is typically the gate capacitance of M1.

Hence, C_h can be expressed as

$$C_h = \alpha C_{ox} W L \quad (57)$$

where α is a bias dependent parameter [18], C_{ox} is the oxide capacitance per unit area, W is M1's channel width and L is M1's channel length. As well, K_1 can be expressed as

$$K_1 = \frac{\mu_n C_{ox} W}{2L} \quad (58)$$

where μ_n is the electron mobility. If (57) and (58) are substituted in the optimum performance measures of the two samplers, the following first-order relationships result:

$$\omega_{\min} \propto \frac{1}{L^2} \quad (59a)$$

$$P_{\text{disp}} \propto C_{ox} \frac{W}{L} \quad (59b)$$

$$\text{SNR} \propto C_{ox} W L. \quad (59c)$$

Combining the three factors indicates that the FOM is independent of C_{ox} , W and L . Hence, the performance of SI circuits is not expected to improve with scaling. Indeed, except for a possible increase in C_{ox} due to thinner oxides, the SNR sets the minimum required area for SI circuits in a manner similar to that of SC circuits [19]. Nevertheless, within the constraints imposed by the technology, varying the length and width of M1 allows the performance of an SI sampler to be tailored to a specific application.

VI. CONCLUSIONS

SI samplers were classified as either triode samplers or saturation samplers, based on their mode of operation. The two sampler types were then analyzed to determine their performance limits and optimum bias levels. Based on the analyzes, the performance of triode mode SI samplers was found to be optimized with the use of the regulated cascode structure. When the triode mode and the saturation mode samplers were compared, the saturation mode was found to offer better overall performance. When biased for optimum performance, both sampler types allow one to tradeoff speed, SNR, and power dissipation to achieve the desired performance mix.

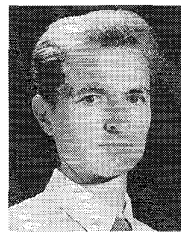
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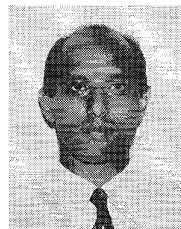
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